RENEWABLE ENERGY SCADA ASSESSMENT

WINDPAT Wind Farm

SCADA Performance Analysis Report

Project	WINDPAT Wind Farm
Asset	15.6 MW wind farm
Analysis period	01 Apr 2024 to 03 Dec 2025
Technology	4 x Nordex N131 3.9 MW
Issued	2026-03-16 15:37 UTC

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HIGHEST-VALUE FINDINGS

Executive Summary

Wind resource capture, fleet availability, and recoverable loss priorities.

Overall assessment

The fleet delivered 59,014 MWh over the analysed period, with a fleet performance index of 72.5% against the derived reference power curve.

Fleet technical availability averaged 88.4%. Estimated availability-led losses account for 9,580 MWh, while residual performance shortfall adds 10,919 MWh.

Annualised recoverable value is €1102.0k/yr (€1,102,005/yr). Annualised from an observed 611-day SCADA period.

Power completeness is 98.9%, wind-speed completeness is 98.9%, and wind-direction completeness is 98.9%. The dataset is sufficiently complete for fleet-level diagnostics.

Fleet performance index

72.5%

Target >= 95%

Fleet availability

88.4%

Target >= 95%

Recoverable loss

€1,102,005/yr

Annualised from an observed 611-day SCADA period.

Recoverable energy

12,245 MWh/yr

Annualised from an observed 611-day SCADA period.

Top Recommended Actions

Priority	Category	Estimated loss (MWh/y)	k€/yr	Action
HIGH	Performance	3,443.9	€309.9k/yr	Perform a targeted power-curve review on LU10, including anemometry, yaw alignment, pitch behaviour, and control settings.
HIGH	Performance	3,263.5	€293.7k/yr	Perform a targeted power-curve review on LU12, including anemometry, yaw alignment, pitch behaviour, and control settings.
HIGH	Performance	3,143.1	€282.9k/yr	Perform a targeted power-curve review on LU11, including anemometry, yaw alignment, pitch behaviour, and control settings.

Key Findings

Primary shortfall driver

Turbines are frequently available but not converting wind resource efficiently, so power-curve underperformance is the leading issue.

Most exposed turbine

LU10 carries the highest annualised recoverable value exposure in the current fleet ranking.

CONSULTANCY DASHBOARD

Performance KPI Dashboard

Core fleet KPIs for the current wind-farm assessment.

KPI interpretation

The dashboard combines operational delivery, data confidence, and annualised value at stake. Annualised from an observed 611-day SCADA period.

Power completeness

98.9%

Target >= 98%

Fleet performance index

72.5%

Target >= 95%

Fleet availability

88.4%

Target >= 95%

Recoverable value

€1102.0k/yr

Annualised from an observed 611-day SCADA period.

Operational Performance

KPI	Value	Reference	Reading
Power completeness	98.9%	>= 98%	SCADA coverage is adequate for fleet diagnostics.
Wind-speed completeness	98.9%	>= 98%	Wind-speed coverage supports power-curve benchmarking.
Fleet performance index	72.5%	>= 95%	Measured energy capture against the fleet reference envelope.

Value And Recoverability

KPI	Value	Reference	Reading
Potential energy	81,423 MWh	Observed period	Upper-envelope production derived from fleet SCADA.
Fleet availability	88.4%	>= 95%	Share of wind-eligible intervals with expected turbine response.
Recoverable loss value	€1,102,005/yr	Annualised from an observed 611-day SCADA period.	Annualised value of current availability and performance shortfall.

PROJECT BASELINE

Site Overview And Technical Scope

Wind-farm scope, asset definition, and engineering method.

Method and asset summary

WINDPAT Wind Farm comprises 4 Nordex N131 3.9 MW turbines (131 m rotor, IEC Class IIA) with a total installed capacity of 15.6 MW.

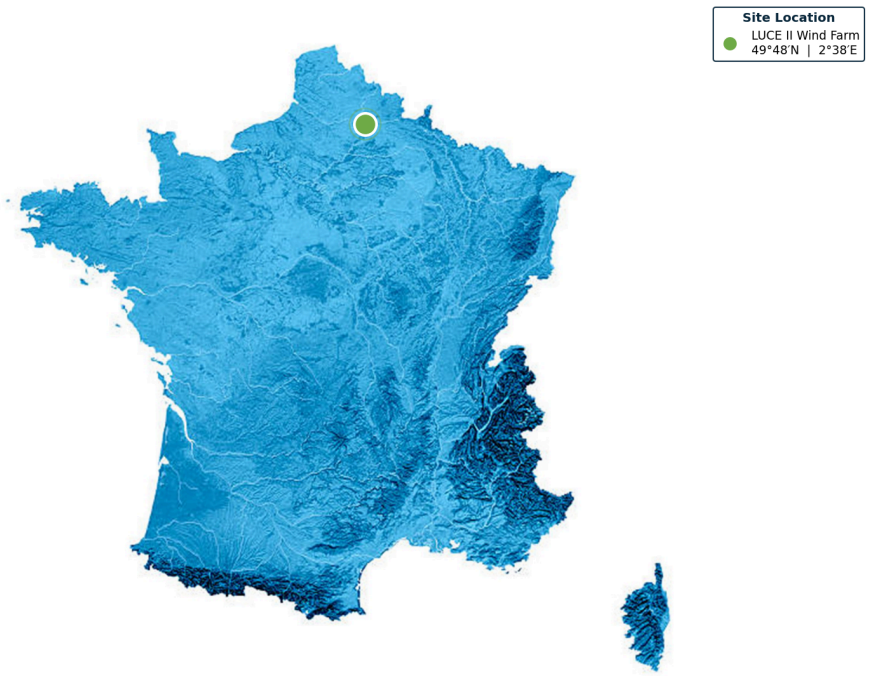
This first-pass WINDPAT assessment uses fleet operation data and manufacturer messages to screen telemetry quality, estimate wind-resource capture, quantify availability-led losses, and prioritise corrective actions.

The Nordex N131 has a design cut-in wind speed of approximately 3 m/s, rated wind speed around 13 m/s, and cut-out at 20 m/s. The fleet-derived reference power curve is benchmarked against this operating envelope; scatter points above 20 m/s with fewer than 12 observations per bin are excluded from the reference to avoid storm-mode artefacts.

Geographic siting is shown from the supplied KMZ marker at 49.8048 N, 2.6393 E.

Installed capacity	Turbines analysed
15.6 MW	**4**

Reporting interval	Tariff assumption
5 min	**€0.09/kWh**



Wind Farm Geographic Location

The location marker is extracted directly from the KMZ supplied with the assessment package.

CONFIGURATION BASIS

Technical Configuration & Analysis Parameters

Asset definition, SCADA basis, and diagnostic assumptions used for this report.

Configuration summary

The table below consolidates the site configuration and the analysis assumptions used to translate raw 5-minute SCADA streams into fleet-level performance and loss indicators.
The turbine layout map shows the GPS position of each wind turbine and the delivery substation (Poste de livraison).
Background: Esri WorldImagery satellite layer.

LUCE II — Turbine Layout

LUCE II — Turbine Layout

GPS coordinates for each turbine and the delivery substation. Esri WorldImagery satellite background.

Site And SCADA Basis

Parameter	Value	Notes
Site name	WINDPAT Wind Farm	Wind farm performance assessment baseline.
Turbine model	Nordex N131 3.9 MW	IEC Class IIA, 131 m rotor diameter, 3,900 kW rated output.
Analysis period	01 Apr 2024 to 03 Dec 2025	Based on the available 5-minute operation export.
Fleet size	4	Turbines included in the fleet-wide assessment.
Rated turbine power	3,900 kW	Nominal 3,900 kW per Nordex N131 specification.

CONFIGURATION BASIS

Technical Configuration & Analysis Parameters cont.

Diagnostic Assumptions

Parameter	Value	Notes
Installed capacity	15.6 MW	Fleet total based on the Nordex N131 3.9 MW rating.
Sampling interval	5 min	Native SCADA export interval.
Primary channels	Power, wind speed, wind direction, nacelle position, rotor speed, generator speed	Used for data-quality and performance screening.
Downtime evidence	Manufacturer message logs and low-output detection	Used together to distinguish low-wind behaviour from operational downtime.
Performance reference	Fleet-derived reference power curve envelope	Constructed from the upper operating envelope across valid wind-speed bins.
Loss valuation	€0.09/kWh	Applied to estimated recoverable energy losses in the action register.

OEM KNOWLEDGE BASE

Turbine Model Intelligence — N131/3900 (Delta4000)

Known failure modes, component weaknesses, and recommended monitoring priorities for the N131/3900 (Delta4000).

Model intelligence summary

The N131/3900 (Delta4000) (3-stage gearbox (medium-speed) + PMSG (AeroTwin drive)) is documented in the 8.2 turbine knowledge base with 7 known failure modes across 6 distinct component groups.

HIGH severity items represent failure modes with documented field frequency and significant revenue impact; they should be cross-checked against the fault message log and SCADA trending data presented in this report.

Design strengths: High annual energy yield at IEC IIA sites due to 131 m rotor and optimised blade profile. | Medium-speed PMSG drivetrain eliminates slip rings and reduces generator maintenance vs DFIG. | Integrated condition monitoring system (CMS) with standard SCADA export. | Service bulletin ecosystem is active with regular OEM updates.

Priority monitoring channels for this model: Pitch fault event rate per turbine per month (target < 1 per turbine/month); Main bearing temperature trend (flag if rising > 2°C/month); Gearbox oil particle count (NAS 7 target in service); Converter coolant system pressure (flag any loss > 0.2 bar/day); Blade leading-edge condition index (drone annual)…

Drivetrain	IEC class
3-stage gearbox	**IIA**

Rotor diameter	HIGH severity issues
131 m	**3** Require attention

N131/3900 (Delta4000) — Delta4000 Platform

Nordex N131 3.9 MW (Delta4000 series). © Nordex SE — used for identification purposes.

OEM KNOWLEDGE BASE

Turbine Model Intelligence — N131/3900 (Delta4000) cont.

Known Issues — N131/3900 (Delta4000)

Component	Known issue	Severity	Recommendation
Pitch system – inverter PCB	Early production (2017–2019) batches of the pitch inverter PCBs exhibited capacitor electrolyte drying, causing pitch axis faults and emergency stops. Service b…	HIGH	Verify service bulletin NX-SB-XXXX application status; replace affected PCBs during next planned maintenance if not already done. Check SCADA for frequency of '…
Main bearing – front main bearing	The N131 uses a 3-point suspension with a single large-diameter main bearing. Micro-pitting observed in units operating in elevated turbulence (TI > 14%) sites.	HIGH	Grease analysis every 6 months; install acoustic emission sensor on main bearing housing; plan inspection at year 5. Turbulence-adaptive pitch scheduling (if av…
Gearbox – planet stage carrier	Planet carrier pin bore wear observed in a proportion of units at year 4–6, leading to gear contact misalignment and accelerated tooth wear on ring gear.	HIGH	Borescope inspection of planet stage at 40,000-hour interval; oil ferrography for metallic particles; budget for gearbox exchange at year 8–10.
Blade – trailing edge bond (root 0–20 m)	Adhesive bond fatigue at the trailing edge panel joint between 5–20 m from root, consistent with high flap-wise bending cycles in turbulent inflow.	MEDIUM	Annual drone inspection focusing on root section trailing edge; early-stage disbonding is repairable in-situ; late-stage requires factory repair.
Blade – leading edge erosion (tip)	Polyurethane leading edge coating erosion accelerates above tip speed 85 m/s in sites with high-frequency rain events, causing AEP losses of 0.5–1.5% per year.	MEDIUM	Apply leading-edge protection (LEP) shells or tape system by year 2–3; annual drone inspection from year 2 onwards.
Converter – water-cooling pump	The Delta4000 uses a liquid-cooled converter; impeller wear and coolant hose micro-cracks have caused unplanned shutdowns.	MEDIUM	Coolant system inspection and pressure test annually; replace pump impellers at 30,000-hour interval; check coolant inhibitor concentration.
Nacelle – vibration alarm sensitivity	Vibration monitoring thresholds set conservatively from factory; nuisance trips from rotor imbalance after blade icing events.	LOW	Calibrate vibration thresholds to site-specific baseline after first winter season; implement blade heating system if ice accretion expected > 100 h/year.

Key Findings

Known weakness — Pitch system – inverter PCB

Early production (2017–2019) batches of the pitch inverter PCBs exhibited capacitor electrolyte drying, causing pitch axis faults and emergency stops. Service bulletin issued.

Known weakness — Main bearing – front main bearing

The N131 uses a 3-point suspension with a single large-diameter main bearing. Micro-pitting observed in units operating in elevated turbulence (TI > 14%) sites.

Known weakness — Gearbox – planet stage carrier

Planet carrier pin bore wear observed in a proportion of units at year 4–6, leading to gear contact misalignment and accelerated tooth wear on ring gear.

TELEMETRY CONFIDENCE

Data Quality

Power, wind-speed, and wind-direction completeness reviewed before interpreting production losses.

Engineering interpretation

Power completeness averages 98.9% across the fleet. The remaining gaps are concentrated in a limited number of turbines rather than a site-wide telemetry outage.

Wind-speed completeness is 98.9% and wind-direction completeness is 98.9%. That is sufficient for reference power-curve benchmarking and directional context.

Turbines showing noticeably lower completeness than the fleet average should be checked first: a persistent gap in power reporting may mask genuine underperformance, while a persistent gap in wind-speed data makes individual power-curve benchmarking unreliable for that unit. Either case should be raised with the SCADA integrator before the next contractual performance review.

Power completeness

98.9%

Target >= 98%

Wind speed completeness

98.9%

Target >= 98%

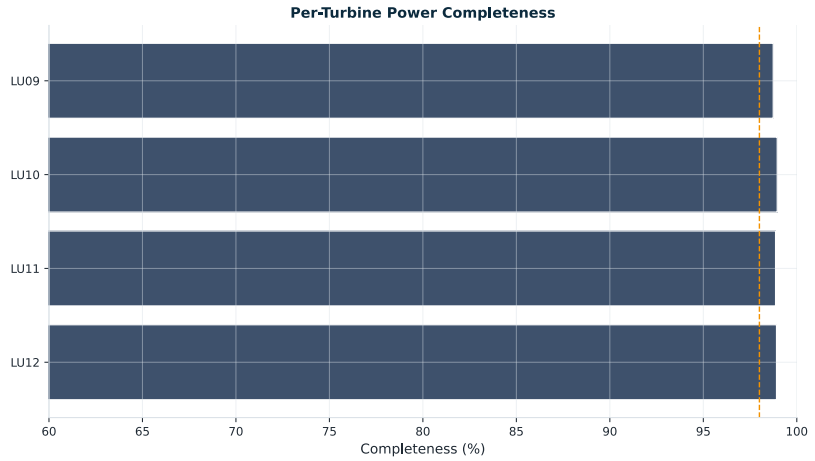
Wind direction completeness

98.9%

Target >= 98%

Valid operating records

697,060



Per-Turbine Power Completeness

The chart highlights turbines where missing power intervals could bias relative performance interpretation.

Key Findings

Data confidence

The available telemetry is strong enough for fleet comparison and loss quantification without material data-confidence caveats.

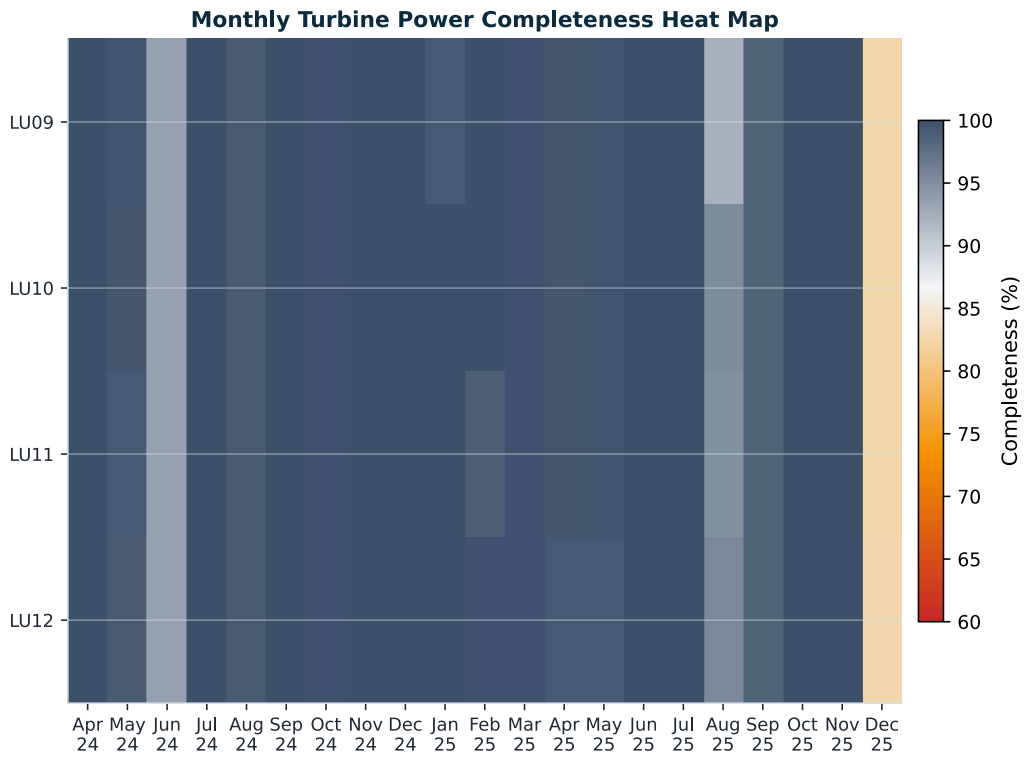
MONTHLY COMPLETENESS

Data Quality Detail

Monthly visibility of turbine-level data gaps.

Monthly interpretation

The monthly heat map is useful for separating persistent channel issues from one-off telemetry outages. Concentrated low-completeness months should be checked before using month-specific performance conclusions for contractual purposes.



Monthly Turbine Power Completeness Heat Map

Recurring low-completeness months are visible by turbine and help distinguish persistent telemetry issues from isolated gaps.

WIND-RESOURCE CAPTURE

Performance Overview

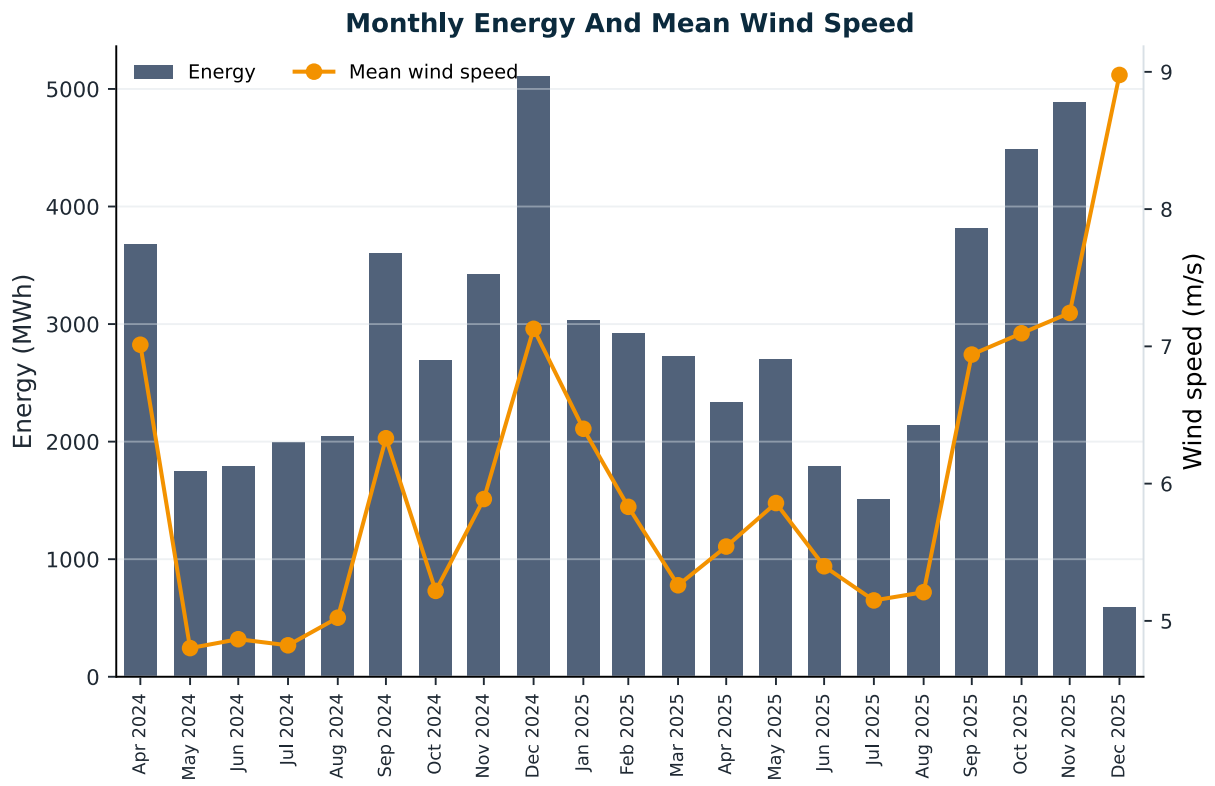
Monthly production and fleet-normalised energy capture.

Performance interpretation

The fleet performance index of 72.5% indicates how much of the derived reference energy envelope was captured. Comparing monthly energy with mean wind speed makes it easier to separate true technical underperformance from simple wind-resource seasonality.

Peak production months (Apr 2024: 5,108 MWh) align with the strongest wind months, confirming the fleet responds broadly to the wind resource. However, months with high wind speed but below-expected energy warrant investigation for curtailment or availability losses.

Note on the final bar in the chart: the SCADA dataset ends on 03 Dec 2025, so the last calendar month is a partial observation (only 3 days). The mean wind speed for that truncated window (9.0 m/s) is measured over the same short period and happened to be unusually high — creating a visible gap against the low partial-month energy. This is a data-boundary artefact, not a technical failure.

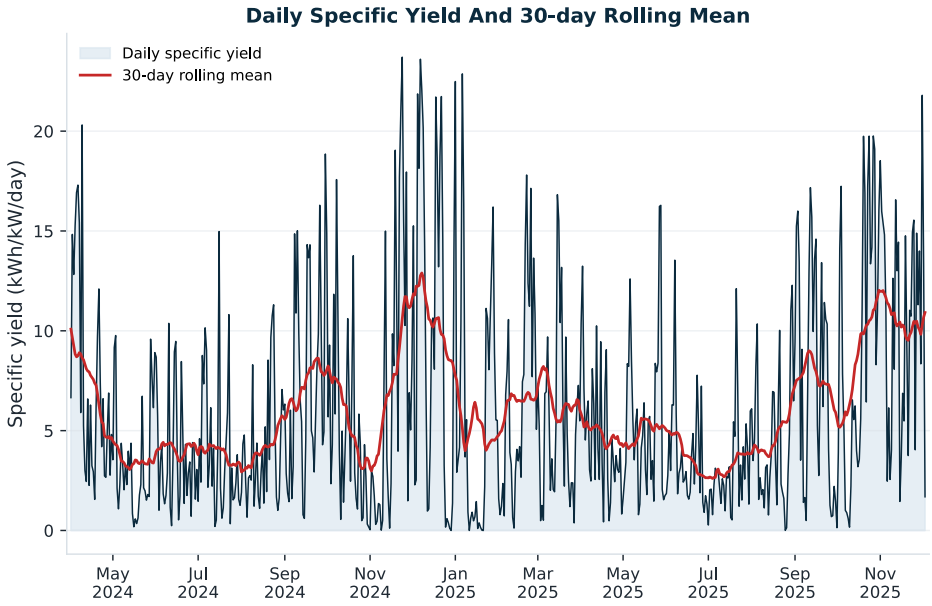


Monthly Energy And Mean Wind Speed

Monthly energy bars are overlaid with mean wind speed so lower production can be compared directly against the wind regime.

WIND-RESOURCE CAPTURE

Performance Overview cont.



Daily Specific Yield And 30-day Rolling Mean

The rolling mean shows whether production weakness is persistent rather than driven by isolated trip days.

Key Findings

Last month artefact

The final month in the chart (Dec 2025) is a partial period ending 03 Dec 2025. The apparent wind-speed/energy gap is a dataset truncation effect — do not interpret as curtailment or underperformance.

RELATIVE RANKING

Fleet Turbine Comparison

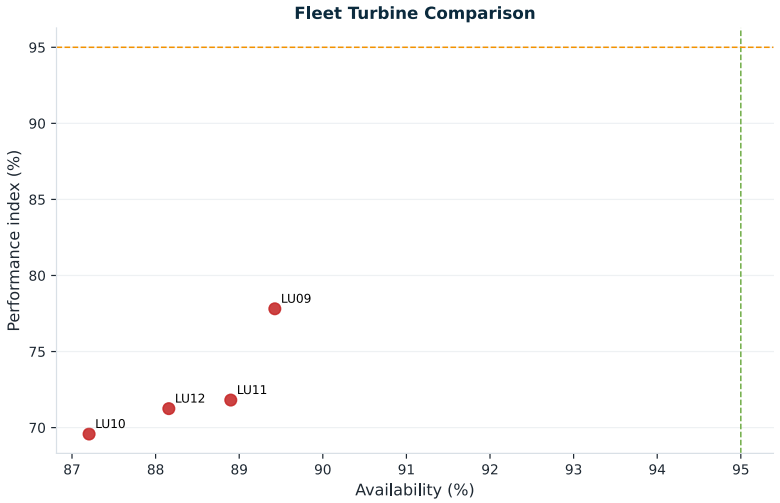
Availability and performance-index comparison across the four turbines.

Fleet interpretation

The scatter separates turbines that are mainly downtime-driven (left side of the chart, low availability) from those that remain available but convert wind resource less efficiently than the fleet envelope (lower PI, high availability).

Turbines in the lower-left quadrant (availability < 95%, PI < 95%) are the highest-priority intervention targets as they carry both an uptime and an efficiency deficit simultaneously. LU10 currently shows the highest annualised recoverable exposure.

Turbines with good availability but low PI often indicate blade pitch mis-calibration, nacelle misalignment, or a deteriorating gearbox — issues that do not trigger hard faults but gradually erode the power curve. These should be scheduled for a blade inspection and pitch audit during the next maintenance window.



Performance Index Versus Availability

Lower-left points warrant immediate intervention because both uptime and energy capture are weak.

Lowest Performing Turbines

Turbine	Availability	Performance index	Recoverable loss	k€/yr
LU10	87.2%	69.6%	3,443.9 MWh/yr	€309.9k/yr
LU12	88.2%	71.2%	3,263.5 MWh/yr	€293.7k/yr
LU11	88.9%	71.8%	3,143.1 MWh/yr	€282.9k/yr
LU09	89.4%	77.8%	2,394.0 MWh/yr	€215.5k/yr

OPERATIONAL CONTINUITY

Availability And Reliability

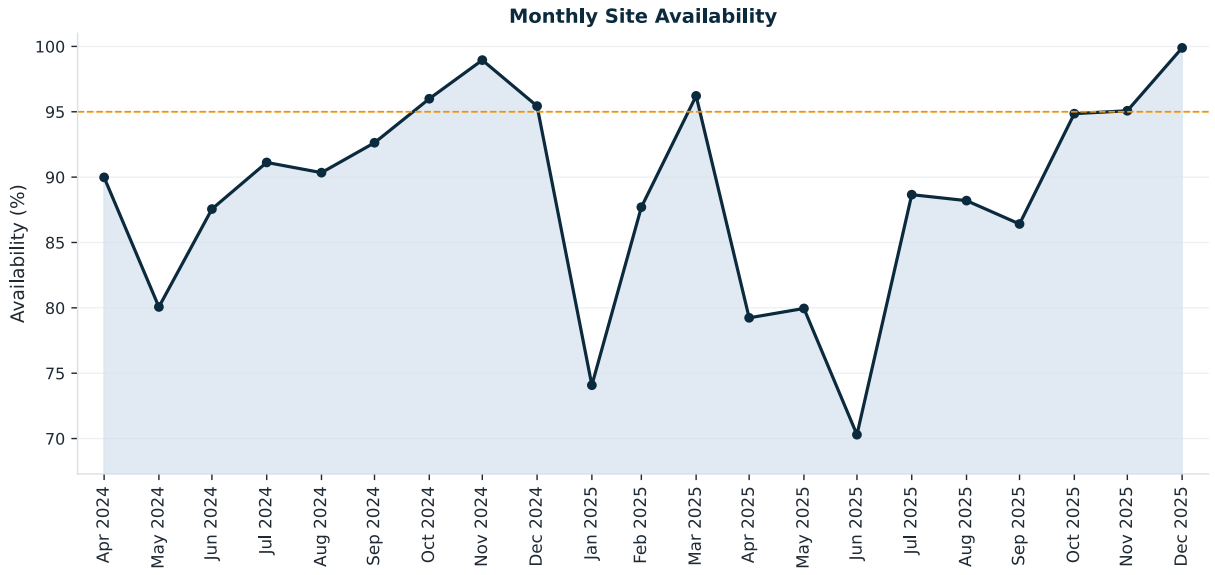
Monthly availability and dominant downtime drivers from the message logs.

Reliability interpretation

Site availability averaged 88.4%. Recurring manufacturer-status messages were screened to separate genuine technical stoppages from low-wind idle conditions.

A fleet availability below 95% during high-wind months has an outsized revenue impact because each lost hour corresponds to near-rated production. Any month where availability drops below 90% should be cross-checked against the fault message logs to confirm whether a single extended outage or repeated short trips drove the deficit.

Persistent recurrence of the same fault family across several turbines usually indicates either a common subsystem weakness or a fleet-wide maintenance practice issue rather than an isolated unit event. Contrarily, single-turbine recurring faults point to unit-specific mechanical or sensor degradation.



Monthly Site Availability

Availability is computed only for wind-eligible intervals, so the metric reflects technical uptime rather than the underlying wind regime.

Lowest Availability / Highest Downtime Units

Turbine	Availability	Downtime	Top fault family
LU10	87.2%	672.7 h	Yaw Wind Speed LowYawEnable
LU12	88.2%	623.2 h	HMI WTG Manual Stop
LU11	88.9%	573.5 h	Yaw Wind Speed LowYawEnable
LU09	89.4%	528.4 h	Yaw Wind Speed LowYawEnable

VALUE AT STAKE

Losses And Recoverability

Potential energy, availability loss, performance loss, and recoverable value.

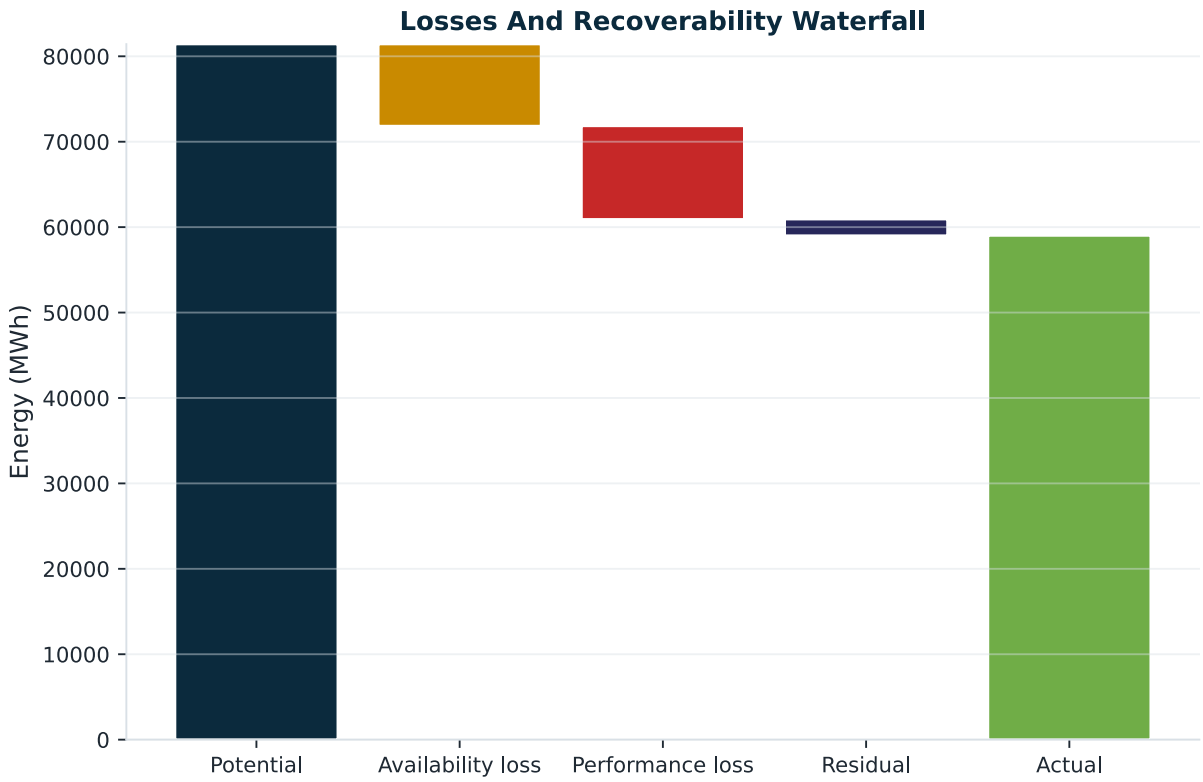
Loss interpretation

Recoverable losses total 20,499 MWh over the analysed period, equivalent to €1102.0k/yr annualised.

Availability loss quantifies periods where the wind resource was present but one or more turbines were effectively unavailable. Performance loss captures sub-envelope operation after excluding those clear downtime periods.

Performance loss is the larger component — turbines are broadly available but are not converting the wind resource at the reference envelope rate. Blade, pitch, and yaw-alignment investigations should take priority.

The waterfall chart makes it easier to communicate the loss structure to asset owners: start from potential energy, subtract availability loss (controllable via maintenance), then performance loss (controllable via optimisation), and the residual is unexplained shortfall that may warrant deeper investigation.

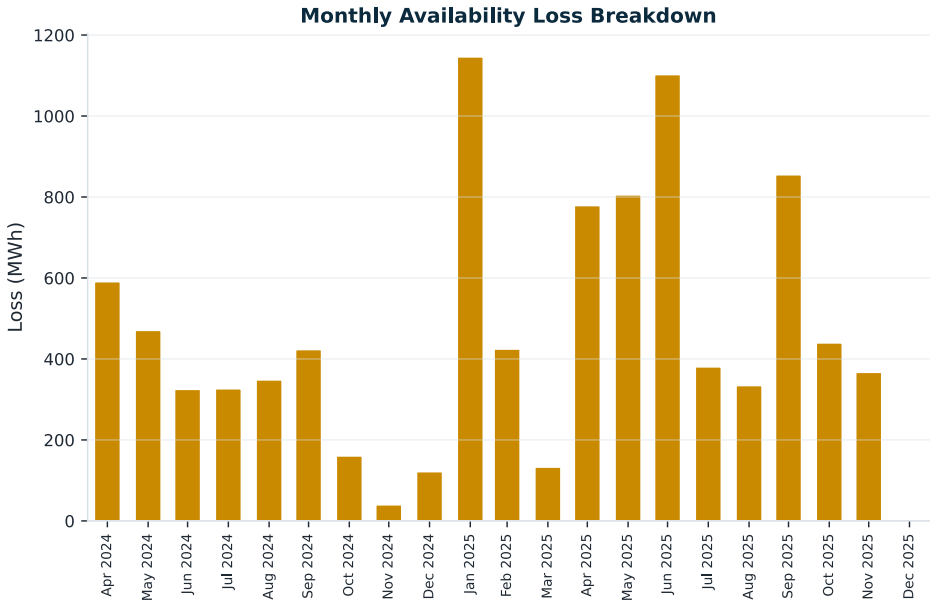


Losses And Recoverability Waterfall

The waterfall starts from fleet potential energy and shows how availability and performance losses reduce realised production.

VALUE AT STAKE

Losses And Recoverability cont.



Monthly Availability Loss Breakdown

This chart highlights which months contributed most to the total availability deficit.

Loss Summary

Metric	Value
Potential energy	81,423 MWh
Actual energy	59,014 MWh
Availability loss	9,580 MWh
Performance loss	10,919 MWh
Recoverable energy (annualised)	12,245 MWh/yr
Recoverable value (annualised)	€1,102,005/yr

CORRECTIVE PRIORITIES

Full Action Punchlist

Client-facing action register ordered by current revenue exposure.

Action interpretation

The punchlist ranks actions by annualised recoverable revenue. Annualised from an observed 611-day SCADA period.
HIGH priority items carry the largest revenue exposure and should be addressed within the current maintenance cycle.
MEDIUM items represent incremental gains that accumulate meaningfully over a 12-month horizon and should be included in the next planned service visit.
Where multiple actions target the same turbine, sequence them to minimise repeated access trips — grouping field work by unit reduces logistics overhead and improves contractor efficiency.

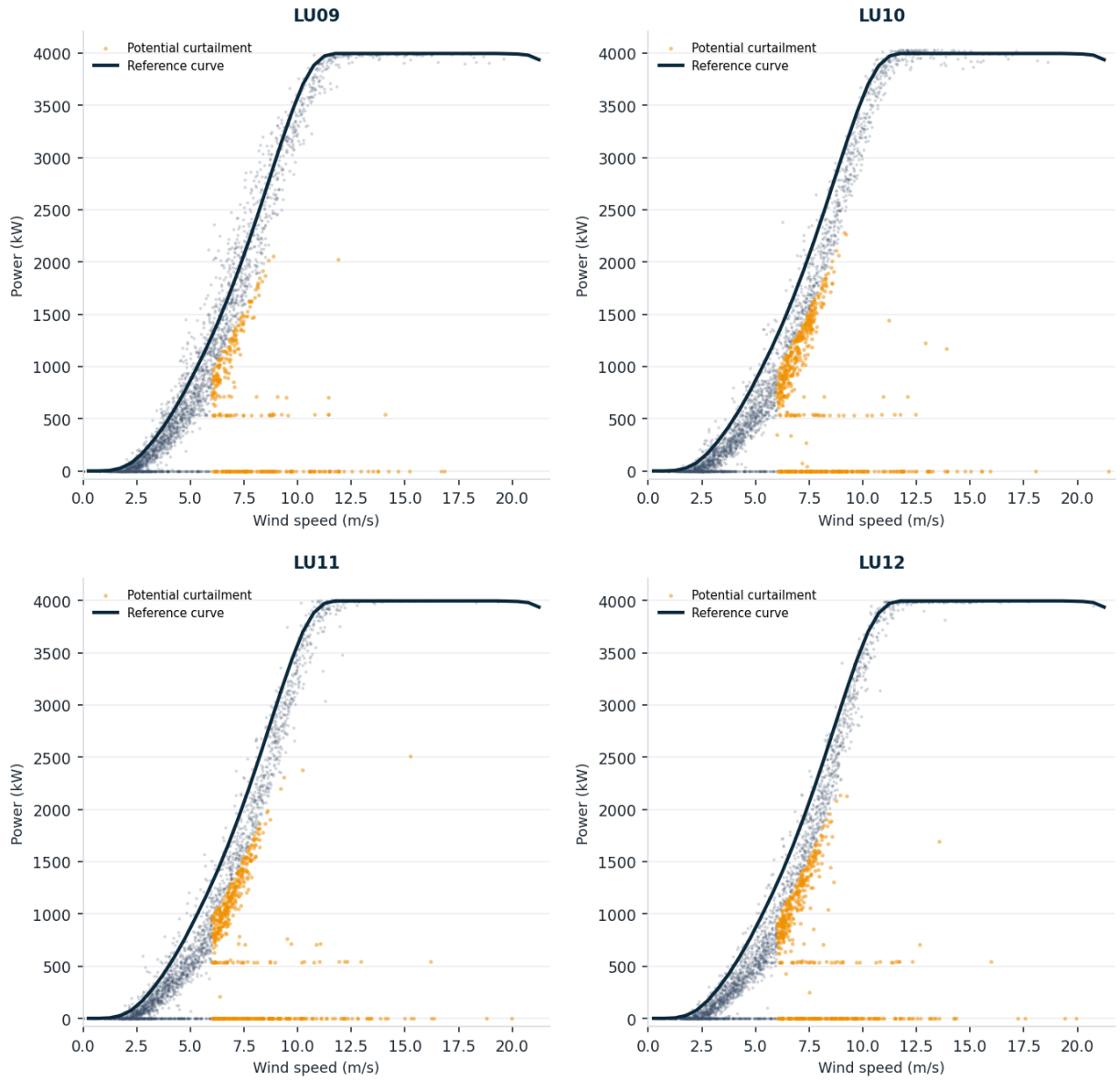
Full Action Punchlist

Priority	Category	Issue	Recommended action	Estimated loss (MWh/y)	k€/yr
HIGH	Performance	LU10 remains available but underperforms the fleet reference power curve.	Perform a targeted power-curve review on LU10, including anemometry, yaw alignment, pitch behaviour, and control settings.	3,443.9	€309.9k/yr
HIGH	Performance	LU12 remains available but underperforms the fleet reference power curve.	Perform a targeted power-curve review on LU12, including anemometry, yaw alignment, pitch behaviour, and control settings.	3,263.5	€293.7k/yr
HIGH	Performance	LU11 remains available but underperforms the fleet reference power curve.	Perform a targeted power-curve review on LU11, including anemometry, yaw alignment, pitch behaviour, and control settings.	3,143.1	€282.9k/yr
HIGH	Availability	LU09 is losing wind-resource capture through repeated downtime during wind-eligible intervals.	Review recurring stop conditions on LU09, inspect the dominant subsystem alerts, and confirm return-to-service latency.	2,394.0	€215.5k/yr
HIGH	Reliability	Yaw Wind Speed LowYawEnable is a recurring downtime driver, particularly on LU10.	Review the underlying subsystem history, confirm spare-parts strategy, and close the recurrence root cause.	1,430.6	€128.8k/yr
HIGH	Reliability	Yaw Wind Speed LowYawEnable is a recurring downtime driver, particularly on LU11.	Review the underlying subsystem history, confirm spare-parts strategy, and close the recurrence root cause.	1,430.6	€128.8k/yr
HIGH	Reliability	Yaw Wind Speed LowYawEnable is a recurring downtime driver, particularly on LU09.	Review the underlying subsystem history, confirm spare-parts strategy, and close the recurrence root cause.	1,430.6	€128.8k/yr

APPENDIX

Appendix - Fleet Power Curve Diagnostics

Scatter of all 10-minute operating points per turbine against the fleet reference curve. Orange points indicate potential curtailment.



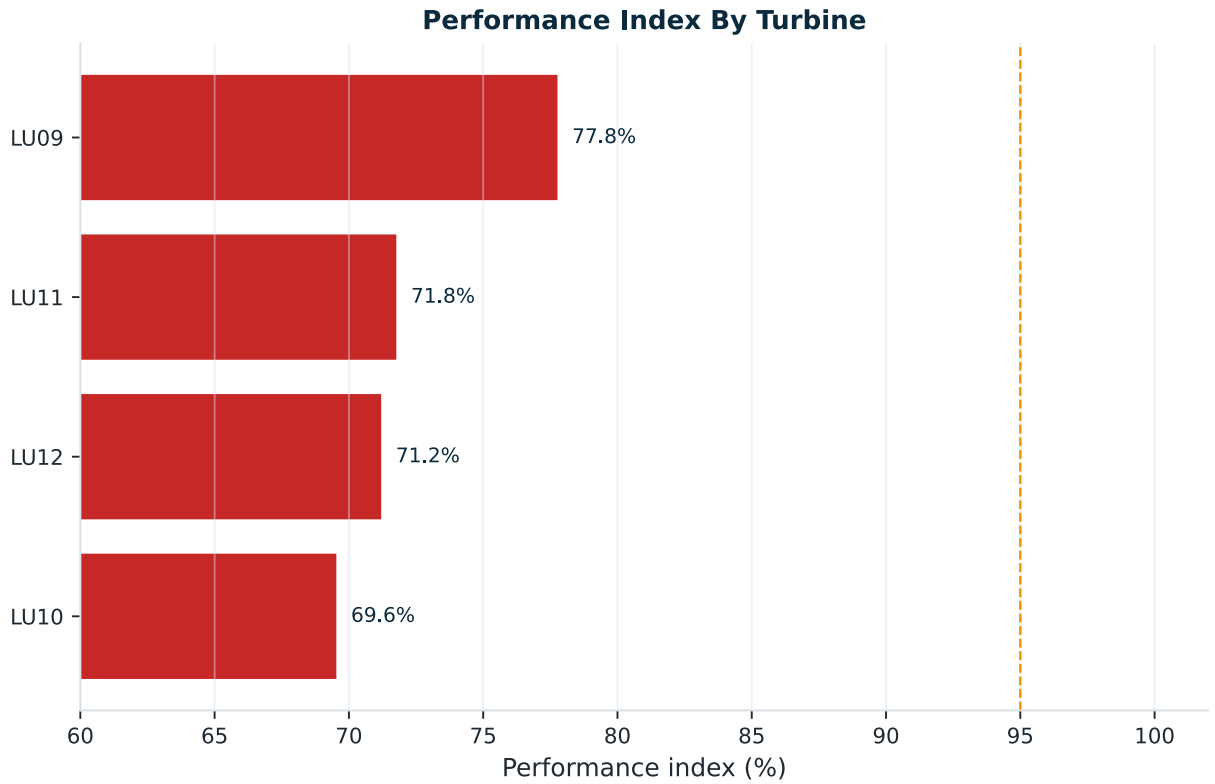
Power Curve Scatter — LU09, LU10, LU11, LU12

Blue = normal operating points. Orange = potential curtailment (wind ≥ 6 m/s, power < 75% of reference).

APPENDIX

Appendix - Turbine Performance Index

Ratio of actual energy to reference-curve potential, annualised over the full analysis period.



Performance Index By Turbine

Turbines below 95% are flagged for investigation. Red < 90%, amber 90–95%, blue ≥ 95%.

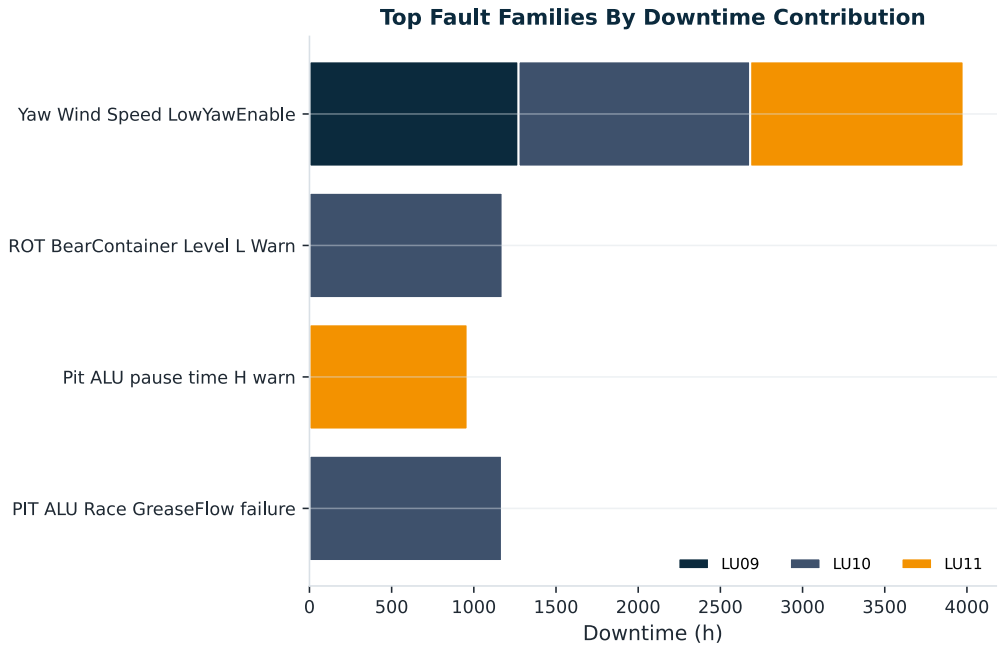
APPENDIX

Appendix - Fault Message Summary

Dominant fault families and downtime contribution by turbine.

Technical Detail

Low-wind and normal-status messages were excluded from the ranked downtime families so the appendix remains focused on actionable operational issues.



Top Fault Families By Downtime Contribution

Downtime hours are grouped by fault family and turbine to highlight common-mode issues.

Top Fault Families

Fault family	Turbine	Count	Downtime	Operational implication
Yaw Wind Speed LowYawEnable	LU10	2,297	1,409.4 h	Inspect the repeated fault family and associated subsystem.
Yaw Wind Speed LowYawEnable	LU11	2,395	1,297.9 h	Inspect the repeated fault family and associated subsystem.
Yaw Wind Speed LowYawEnable	LU09	2,311	1,272.2 h	Inspect the repeated fault family and associated subsystem.
ROT BearContainer Level L Warn	LU10	92	1,175.8 h	Inspect the repeated fault family and associated subsystem.
PIT ALU Race GreaseFlow failure	LU10	53	1,171.0 h	Inspect the repeated fault family and associated subsystem.
Pit ALU pause time H warn	LU11	33	961.9 h	Inspect the repeated fault family and associated subsystem.